NEON SPIRE

GAME DESIGN DOCUMENT

A cylindrical tower-climbing autorunner

Inspired by Nebulus (1987) • Reimagined in Three.js

GameVolt.io

Version 1.0 – March 2026

Status: Pre-production / Concept

1. GAME OVERVIEW

Parameter	Details
Title	Neon Spire (working title)
Genre	Vertical autorunner / platformer
Platform	Browser (desktop + mobile) via GameVolt.io
Engine	Three.js (WebGL)
Inspiration	Nebulus / Tower Toppler (Hewson, 1987)
Target audience	Casual-to-core gamers, retro enthusiasts, synthwave fans
Session length	2–5 minutes per tower attempt
Monetization	Free-to-play, ad-supported (GameVolt standard)

1.1 Elevator Pitch

Neon Spire is a vertical autorunner where the player climbs the outside of a glowing cylindrical tower, jumping between platforms that spiral upward, dodging enemies and hazards. The tower rotates automatically – the player controls jump timing, direction changes, and special abilities. Think Nebulus meets Synthwave meets modern browser gaming.

1.2 Unique Selling Points

- **Cylindrical 3D gameplay:** A nearly extinct mechanic reborn in modern WebGL – platforms wrap around a real 3D cylinder, creating natural occlusion and reveal moments.
- **Autorunner meets platformer:** The tower rotates automatically but the player has full control over jumps, direction reversal, and abilities – more agency than typical autorunners.
- **Synthwave architecture:** The tower itself is the visual centerpiece – a glowing neon structure against a deep space backdrop, not just a backdrop but a character.
- **Dead genre, fresh execution:** No modern game occupies this niche. Cylindrical platforming is effectively a blue ocean.

2. CORE MECHANICS

2.1 The Cylinder

The game world is a vertical cylinder. All gameplay takes place on its outer surface. The player's position is defined by two values: angle (theta, 0–360° around the cylinder) and height (Y position). Platforms, enemies, and collectibles are all placed in this polar coordinate space.

The cylinder rotates continuously (or the player runs around it – visually equivalent). The camera is fixed on the outside, tracking the player, so platforms naturally appear from one side and disappear behind the tower on the other. This creates a built-in fog-of-war effect without any artificial fog.

2.2 Player Controls

Input	Action	Notes

Tap / Space	Jump	Variable height based on press duration
Swipe left-right / A-D	Reverse direction	Player reverses running direction on the cylinder surface
Swipe up / W	High jump	Costs energy, reaches platforms 2 levels up
Tap during jump	Double jump	Unlocked ability, small directional adjustment
Hold down / S	Duck / slide	Pass under low obstacles, faster descent on slopes
Double-swipe / Shift	Dash	Burst of speed forward along the cylinder surface. Unlocked from Tower 3. Cooldown-based, i-frames during dash. Costs energy.

2.3 Auto-rotation

The player character runs automatically around the cylinder at a base speed. Speed increases gradually per tower level and can be temporarily boosted by pickups. The player cannot stop moving – the core challenge is timing jumps and direction changes to navigate the spiraling platforms.

Direction reversal is instant but has a short animation. Strategic direction changes are key to advanced play – sometimes running “backwards” to catch a platform you just passed is the optimal move.

2.4 Dash Mechanic

Unlocked from Tower 3 (Void Pillar) onward, the dash gives the player a short burst of speed along the cylinder surface in the current running direction. The dash is activated by a double-swipe on mobile or Shift on keyboard. During the dash, the player has invincibility frames (i-frames), allowing them to pass through enemies and hazards unharmed.

The dash operates on a cooldown (3 seconds) and costs energy, creating a risk/reward dynamic: use it to escape danger, skip past a tricky hazard section, or reach a distant platform – but if you waste it, you’re vulnerable until it recharges. The dash can be combined with a jump for a “dash-jump” that covers significant horizontal arc distance, essential for some late-game platform gaps.

Visually, the dash leaves a bright afterimage trail along the cylinder surface and triggers a brief speed-line effect. The player character becomes a streak of light for the dash duration (~0.3 seconds), making it feel impactful and rewarding.

2.5 Vertical Progression

The camera follows the player upward. Falling below the visible area means death (or losing a life). Platforms are arranged in spiral patterns that require the player to jump upward while the cylinder rotates. Some platforms are stationary, others move, crumble, or bounce the player.

2.6 Platform Types

Platform	Behavior	Visual
Solid	Static, reliable	Cyan glow, steady
Crumbling	Breaks 0.5s after landing	Flickering, unstable glow
Bouncy	Launches player extra high	Magenta pulse on contact
Moving	Oscillates horizontally (around cylinder)	Trailing afterglow

Phasing	Fades in/out on a timer	Ghost-like transparency cycle
Conveyor	Pushes player in one direction	Animated arrow pattern
Destructible	Can be destroyed by enemies or player ability	Cracked glow texture

3. ENEMIES & HAZARDS

3.1 Enemy Types

Enemy	Behavior	Counter
Sentinel	Patrols a platform back and forth	Jump over or reverse direction to avoid
Orbiter	Circles the tower at a fixed height	Time your vertical position to pass safely
Dropper	Falls from above when player is below	Keep moving – audio cue warns of drop
Zapper	Stationary, emits periodic beam across arc	Time passage between zap cycles
Swarm	Group of small drones that track loosely	Outrun them – they're slower than the player

3.2 Environmental Hazards

- Laser gates – Horizontal beams that block sections of the tower, on/off cycle
- Falling debris – Blocks that tumble down the tower surface, telegraphed by shadow
- Wind zones – Push the player in the rotation direction, affecting jump arcs
- Dark zones – Sections where neon glow dims, reducing visibility to immediate surroundings
- EMP pulses – Temporarily disable the player's HUD and ability indicators

4. PROGRESSION & STRUCTURE

4.1 Tower Structure

Each tower is a self-contained level, divided into vertical sections (zones). A tower consists of 4–6 zones, each with a distinct visual theme variation and increasing difficulty. Completing a tower (reaching the top) unlocks the next tower.

4.2 Zone Progression

Zone	Theme	New Element Introduced
1 – Foundation	Standard cyan/blue neon	Basic platforms, Sentinels
2 – Interference	Added magenta interference lines	Crumbling platforms, Orbiters

3 – Storm	Electrical arcs, flickering lights	Zappers, wind zones, moving platforms, dash unlocked
4 – Void	Dark sections, reduced glow	Dark zones, Droppers, phasing platforms
5 – Core	Intense glow, high speed	All elements combined, increased speed
6 – Apex (Boss)	Unique per tower	Boss encounter at tower summit

4.3 Towers (Worlds)

The game launches with 5 towers, each with a unique visual identity built on the synthwave foundation:

- **Tower 1 – Signal Spire:** Classic cyan/magenta, tutorial tower, forgiving timing.
- **Tower 2 – Chrome Needle:** Silver/white metallic look, introduces faster rotation.
- **Tower 3 – Void Pillar:** Deep purple/black, heavy use of dark zones and phasing. Dash mechanic introduced here – required to navigate wider platform gaps.
- **Tower 4 – Solar Furnace:** Orange/gold neon, heat distortion effects, debris hazards.
- **Tower 5 – The Architect:** All colors, randomized zone order, ultimate challenge.

4.4 Scoring System

- Height bonus – Points per vertical meter climbed
- Speed bonus – Multiplier for maintaining momentum without stopping/reversing
- Collectible orbs – Scattered on and between platforms (10/50/100 point variants)
- Clean zone bonus – Complete a zone without taking damage
- Tower completion time – Bonus for fast completion, feeds into GameVolt leaderboard

4.5 GameVolt Integration

Full GameVolt SDK integration:

- Global leaderboards per tower (score + time)
- Cloud saves for progression (unlocked towers, best scores)
- Trophy/achievement system (e.g., “Complete Tower 1 without reversing”, “Collect all orbs in a zone”)
- Guest play supported – login required only for cloud saves and leaderboards

5. VISUAL DESIGN

5.1 Art Direction

The visual style is “architectural synthwave” – not retro pixel art with neon slapped on, but a coherent vision where the tower is a monumental neon structure floating in a void. Think Tron Legacy meets brutalist architecture meets a music visualizer.

5.2 Key Visual Elements

- **The Tower:** Semi-transparent cylinder with visible internal structure (wireframe glow). Platforms are solid ledges that protrude from the surface. The tower pulses subtly with the music.

- **Background:** Deep space gradient (dark blue to black) with a distant grid plane below. Stars and distant geometric shapes. Parallax layers for depth.
- **Player Character:** Small, glowing humanoid silhouette with a bright trail. Minimal detail – the focus is on movement and the tower. Color shifts based on current ability/state.
- **Lighting:** UnrealBloomPass for all neon elements. Platforms cast subtle glow on the tower surface. Enemies have distinct color-coded glows (red for damage-dealing, yellow for warning).
- **Particles:** Sparks on landing, trail particles while running, burst effects on collectibles, debris on crumbling platforms.

5.3 Camera

Third-person, fixed angle from outside the tower. The camera tracks the player vertically and slightly orbits to keep the player in a comfortable screen position. The tower fills roughly 40–50% of the screen width, giving a clear view of upcoming platforms and hazards.

Optional: Slight camera rotation on direction change for dynamic feel. Zoom-out during boss encounters for dramatic effect.

6. AUDIO DESIGN

6.1 Music

Synthesized synthwave/retrowave soundtrack using Web Audio API, consistent with the Gravity Well approach. Each tower has a unique track that builds in intensity per zone. BPM-synced visual pulses on the tower geometry.

Music layers crossfade based on gameplay state: calm exploration layers, intense combat/speed layers, and a victory sting on tower completion.

6.2 Sound Effects

All SFX generated via Web Audio synthesis (no external audio files):

- Jump – Short rising synth chirp
- Land – Soft thud with reverb
- Direction change – Quick whoosh/sweep
- Dash – Aggressive acceleration burst with electric crackle, doppler tail-off
- Collectible – Bright arpeggio note (pitch varies)
- Enemy proximity – Low pulsing hum (spatial audio, louder when closer)
- Crumbling platform – Crackling/breaking sound
- Death – Descending tone + static burst
- Zone transition – Rising sweep + bass drop

7. TECHNICAL ARCHITECTURE

7.1 Core Tech

Component	Technology
Rendering	Three.js (WebGL 2.0)
Geometry	CylinderGeometry for tower, custom BufferGeometry for platforms

Post-processing	EffectComposer: UnrealBloomPass, optional ChromaticAberration
Physics	Custom polar coordinate collision (no physics engine needed)
Audio	Web Audio API (synthesized music + SFX, Gravity Well approach)
State management	Simple state machine (menu, playing, paused, dead, victory)
Backend	Supabase via GameVolt SDK (auth, saves, leaderboards, trophies)

7.2 Coordinate System

All game objects use polar coordinates internally:

- theta (0–360°) – Position around the cylinder circumference
- y – Height on the cylinder
- r – Radial offset (for objects that float slightly off the surface)

Conversion to Three.js world coordinates: x = r * cos(theta), z = r * sin(theta), y = y. This makes collision detection trivial – compare theta ranges and y ranges rather than full 3D intersection.

7.3 Level Data Format

Towers are defined as JSON data: an array of platform objects with type, theta, y, width (angular span), and optional behavior parameters. Enemy placements similarly. This allows easy procedural generation for endless/challenge modes later.

7.4 Performance Targets

- 60 FPS on mid-range hardware (M1 Mac, modern integrated GPU)
- 30 FPS minimum on mobile browsers
- Draw call optimization: instanced rendering for repeated platform geometry
- LOD: Platforms on the far side of the cylinder rendered at lower detail
- Bloom pass at half resolution for performance

8. MOBILE & INPUT CONSIDERATIONS

The game should work on both desktop and mobile from day one:

- Desktop: Keyboard (Space/W/A/D/S) + optional mouse click for jump
- Mobile: Touch zones – left side tap = reverse, right side tap = jump, swipe up = high jump, swipe down = duck
- Gamepad: Supported via Gamepad API (A = jump, triggers = reverse, left stick for high jump/duck)

The cylindrical gameplay actually maps very well to mobile – the limited input vocabulary (jump, reverse, duck) means touch controls can be clean and uncluttered.

9. MVP SCOPE

For the first playable prototype:

9.1 Must Have (Prototype)

- Rotating cylinder with player character running on surface

- 5–8 platform types placed on cylinder
- Jump + direction reversal controls
- One complete tower with 3 zones (tutorial, intermediate, challenge)
- Basic synthwave visual style with bloom
- Score tracking
- Death + restart loop

9.2 Should Have (v1.0 Launch)

- 3 complete towers
- All enemy types
- Full synthesized audio (music + SFX)
- GameVolt SDK: leaderboards, cloud saves, trophies
- Mobile touch controls
- Zone transition effects

9.3 Nice to Have (Post-launch)

- Endless/procedural mode
- Boss encounters
- Additional towers (community favorites / seasonal)
- Unlockable player skins/trails
- Daily challenge tower (seeded generation)
- Replay/ghost system

10. RISKS & MITIGATIONS

Risk	Impact	Mitigation
Cylindrical controls feel disorienting	High	Gradual speed ramp in Tower 1, visual anchor points on cylinder, subtle rotation indicators
Mobile touch precision for jumping	Medium	Generous input windows, aim assist on platforms, coyote time
Bloom performance on low-end devices	Medium	Configurable quality settings, half-res bloom, disable on low-end
Level design for wrapping geometry	Medium	JSON-based level format allows rapid iteration and playtesting
Motion sickness from rotation	Low-Med	Fixed camera angle (no player-controlled rotation), smooth interpolation, optional reduced motion mode

11. REFERENCES & INSPIRATION

- **Nebulus / Tower Toppler (1987):** Core cylindrical tower-climbing mechanic. Proof that the concept works and is compelling.
- **Vampire Survivors (2022):** Proof that reviving a dead genre with modern polish can be massively successful.

- **Celeste (2018):** Precision platformer with tight controls and clear visual language for hazards.
- **Tron Legacy (2010):** Visual reference for architectural neon aesthetic.
- **Gravity Well (GameVolt):** Internal reference for Web Audio synthesis approach and synthwave visual style.